

Ideal and Non-ideal solution

21. The solution which shows negative or positive deviation by Raoult's law, is called
- Ideal solution
 - Real solution
 - Non-ideal solution
 - Colloidal solution
22. Which of the following does not show positive deviation from Raoult's law
- Benzene-Chloroform
 - Benzene-Acetone
 - Benzene-Ethanol
 - Benzene-Carbon tetrachloride
23. Which of the following mixture shows positive deviation by ideal behaviour
- $\text{CHCl}_3 + (\text{CH}_3)_2\text{CO}$
 - $\text{C}_6\text{H}_6 + \text{C}_6\text{H}_5\text{CH}_3$
 - $\text{H}_2\text{O} + \text{HCl}$
 - $\text{CCl}_4 + \text{CHCl}_3$
24. Which property is not found in ideal solution
- $P_A \neq P_A^0 \times X_A$
 - $\Delta H_{\text{mix}} \neq 0$
 - $\Delta V_{\text{mix}} \neq 0$
 - All of these
25. Which of the following is not correct for ideal solution
- $\Delta S_{\text{mix}} = 0$
 - $\Delta H_{\text{mix}} = 0$
 - It obeys Raoult's law
 - $\Delta V_{\text{mix}} = 0$
26. Which of the following does not show negative deviation from Raoult's law
- Acetone-Chloroform
 - Acetone-Benzene
 - Chloroform-Ether
 - Chloroform-Benzene
27. A mixture of benzene and toluene forms
- An ideal solution
 - Non-ideal solution
 - Suspension
 - Emulsion
28. Which of the following is an ideal solution
- Water + ethanol
 - Chloroform + carbon tetrachloride
 - Benzene + toluene
 - Water + hydrochloric acid
29. When ethanol mixes in cyclohexane; cyclohexane reduces the intermolecular forces between ethanol molecule. In this, liquid pair shows



- (a) Positive deviation by Raoult's law
 - (b) Negative deviation by Raoult's law
 - (c) No deviation by Raoult's law
 - (d) Decrease in volume
30. Liquids *A* and *B* form an ideal solution
- (a) The enthalpy of mixing is zero
 - (b) The entropy of mixing is zero
 - (c) The free energy of mixing is zero
 - (d) The free energy as well as the entropy of mixing are each zero

